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PRAYCG Workflow: From Stimulus Selection to Finished Analysis

Document version: v1.0

Target audience: Windows users who are only mildly familiar with Python, OpenBCI, LabRecorder, and PsychoPy.

Software target: PRAYCG Control Center v0.5 self-contained bundle.

1. One-page overview

A complete PRAYCG run has eight major stages:

1. Select a legal, suitable master stimulus.
2. Add the master stimulus to the PRAYCG Stimulus Library.
3. Run MediaPrep to generate Target, Override, Control, cue schedule, and QC files.
4. Verify media, audio, cue, anchor, and ALS timing quality.
5. Start acquisition streams: OpenBCI/BrainFlow, ALS/PT19 if used, Polar H10, Vernier respiration.
6. Open LabRecorder and record all LSL streams into one XDF.
7. Launch PRAYCG2.0 from PsychoPy Coder and complete the protocol.
8. Run Master Comprehensive Analysis, Visualizer, and Offline Interpreter.

The Control Center is the front door. It launches and organizes the tools. It is not a replacement for PsychoPy, LabRecorder, OpenBCI drivers, or manual QC.

2. What counts as a good PRAYCG stimulus

A good stimulus should be:

- [] legally usable by the person running the study
- [] not redistributed publicly if copyrighted
- [] 3 to 5 minutes long when possible
- [] emotionally or narratively meaningful enough to plausibly matter
- [] clear enough that dialogue and visual events can be understood
- [] not so loud, flashing, or chaotic that sensory overload dominates
- [] visually stable enough that number cues remain readable
- [] suitable for the participant's comfort and consent

Avoid:

- [] poor audio
- [] hard subtitles or platform watermarks unless intentionally documented
- [] extreme flashing/strobe content
- [] scenes that depend on unshared background context
- [] scenes too familiar to the participant if the goal is novelty
- [] scenes too personally disturbing without proper safeguards

A stimulus can be emotionally powerful but still scientifically poor if the control cannot be made valid, the audio is unintelligible, or the anchors cannot be locked.

3. Create the PRAYCG working folders

Recommended folder root:

C:\PRAYCG\

Suggested structure:

C:\PRAYCG\
 stimuli\
 runs\
 analysis\
 logs\
 config\

For one stimulus:

C:\PRAYCG\stimuli\MyStimulus\
 master\
 mediaprep\
 anchors\
 qc\

For one run:

C:\PRAYCG\runs\MyStimulus_Run1\
 acquisition\
 runner_logs\
 inputs_snapshot\
 notes\

For one analysis output:

C:\PRAYCG\analysis\MyStimulus_Run1_MasterComprehensive\
 tables\
 reports\

figures\
visualizer\

4. Add a master stimulus in the Control Center

1. Open PRAYCG Control Center v0.5.
2. Go to **Settings**.
3. Click **Use Bundled Paths**.
4. Click **Auto-find ALL**.
5. Go to **Stimulus Library**.
6. Click **Add Master Stimulus**.
7. Choose your master .mp4.
8. Let the Control Center validate the video.

The Control Center should copy the file into the stimulus library and write a validation report.

Expected result:

C:\PRAYCG\stimuli\<stimulus_id>\master\stimulus_master.mp4

C:\PRAYCG\stimuli\<stimulus_id>\master\master_media_validation_report.json

C:\PRAYCG\stimuli\<stimulus_id>\master\master_sha256.txt

If validation warns about missing audio, bad duration, unreadable video, or audio/video mismatch, fix the source file before continuing.

5. Run MediaPrep

1. In Control Center, select your stimulus.
2. Click **Launch MediaPrep**.
3. In MediaPrep, select the master .mp4.
4. Choose or confirm project name.
5. Generate the prepared videos and cue schedule.
6. Enable StimulusFingerprint / CET-EET regressors if available.
7. Save outputs to the stimulus folder.

MediaPrep should generate files similar to:

stimulus_target_cued_*.mp4

stimulus_override_cued_*.mp4

stimulus_control_cued_phase_scrambled_*.mp4

cue_schedule_*.json

cue_schedule_*.csv

qc\stimulus_exogenous_regressor_frame_all_conditions.csv

qc\cet_regressors_all_conditions.csv
qc\stimulusfingerprint_cet_eet_manifest.json

The important method rule is:

Target and Override should be the same cue-embedded stimulus.

Only participant instructions should differ.

Control should be generated from the cue-embedded Target.

This protects the Target-vs-Override comparison and ensures cue timing/energy are represented in the Control branch.

6. Check MediaPrep outputs

After MediaPrep finishes, in Control Center click:

Scan Selected Stimulus

You want:

Target: FOUND

Override: FOUND

Control: FOUND

Cue JSON: FOUND

Cue CSV: FOUND

QC folder: FOUND

Anchor file: LOCKED FOUND or DRAFT FOUND

If Target and Override are not both found, do not run.

If the cue schedule is missing, do not run.

If only an estimated or draft anchor file exists, label the run as pilot-only unless you frame-lock it before acquisition.

7. Manual media QC

Before any real recording, watch or listen to the prepared files.

Visual QC

Watch Target, Override, and Control.

Pass criteria:

[] Target and Override play correctly.

[] Target and Override are visually identical apart from intended instructions outside

the video.

☐ Number cues are visible and readable.

☐ No unreadable source stamp, watermark, logo, or subtitle artifact remains unless documented.

☐ Any visual cleanup happened before Target, Override, and Control were generated.

☐ Control was generated from the cleaned cue-embedded Target.

Control audio QC

Listen to the full phase-scrambled Control using the same speaker/headphone settings as the protocol.

Pass criteria:

☐ no recognizable spoken words

☐ no intelligible sentence fragments

☐ no distinct melody that carries narrative recognition

☐ loudness/envelope broadly tracks the original

☐ control sounds like speech-shaped or scene-shaped noise, not understandable dialogue

If recognizable speech remains, label the run:

PILOT ONLY

or regenerate the Control.

8. Create or verify anchors

Anchors are predeclared moments in the stimulus that the analysis will later test.

Anchor status levels:

DRAFT_NULL_TIMES:

structure only; not usable as-is.

ESTIMATED:

can be loaded for pilot runs; not strict confirmation-grade.

LOCKED:

exact rendered_time_sec values verified on the final MediaPrep Target MP4 before acquisition.

For confirmation-grade A-MRED attempts:

1. Open final rendered Target MP4.

2. Find each anchor event.

3. Record exact rendered_time_sec.

4. Enter times into the anchor JSON.
5. Save as *_LOCKED.json.
6. Load the LOCKED JSON in PRAYCG2.0 before acquisition.

A runner-loaded estimated file is useful, but it is not strict confirmation-grade.

9. Prepare hardware

OpenBCI Cyton + Daisy

- ☐ Cyton + Daisy board charged or powered.
- ☐ Dongle inserted.
- ☐ Board recognized by Windows.
- ☐ COM port identified.
- ☐ OpenBCI GUI test stream passes.
- ☐ OpenBCI GUI closed before BrainFlow starts.

Gelfree/gelless cap

- ☐ Cap fitted correctly.
- ☐ Electrodes placed according to PRAYCG16 mapping.
- ☐ Reference and ground connected correctly.
- ☐ Hair moved away from contacts.
- ☐ Contact/signal quality checked.
- ☐ Channel map physically verified.

ALS/PT19 light sensor

- ☐ Sensor held firmly over timing square or runner-level test pulse region.
- ☐ Opaque shroud blocks room light.
- ☐ Cable strain relieved.
- ☐ ALS signal visible in the OpenBCI AUX/analog stream.
- ☐ White pulse produces strong rise and return to baseline.
- ☐ Control, Target, and Override all pass pulse test.

A 3D-printed ALS holder is recommended because sensor position and shrouding matter more than just pulse length.

Polar H10

- ☐ Strap wet and fitted.
- ☐ Bluetooth on.
- ☐ Not connected to another phone app.
- ☐ PolarHRV stream appears in LSL.
- ☐ RR intervals are being pushed.

Vernier Respiration Belt

- [] Belt fitted around torso.
 - [] USB preferred for stability.
 - [] VernierRespirationBelt stream appears in LSL.
 - [] force_N channel changes with breathing.
-

10. Start acquisition streams

Open Control Center and go to **Acquisition**.

Recommended stream-start order:

1. Start Polar H10 RR -> LSL.
2. Start Vernier Respiration USB -> LSL.
3. Start BrainFlow EEG Only or BrainFlow EEG + ALS.
4. Check visible LSL streams.
5. Run Signal Quality / Contact Index.
6. Run ALS pulse test if using ALS/PT19.
7. Open LabRecorder.

Expected stream names may include:

PolarHRV
VernierRespirationBelt
obci_eeg1
OpenBCIStatusMarkers
praycg_stasismarkers_...

Do not start LabRecorder until you have the required streams online or you know which late stream you are waiting for.

11. Use LabRecorder

1. Open LabRecorder from Control Center.
2. Click **Update**.
3. Check the boxes for required streams.
4. Set the save folder to:
C:\PRAYCG\runs\<run_id>\acquisition\

5. Use a clear filename, such as:
sub-P001_ses-S001_task-PRAYCG_run-001_eeg.xdf
6. When the PRAYCG marker stream is online, click **Start**.
7. Confirm the file size is increasing.

8. Do not stop LabRecorder until the entire PRAYCG protocol is finished.
9. Click **Stop** after the final report and protocol end.

If LabRecorder is missing a stream, click **Update** again. If a required stream remains missing, label the run as pilot or abort and fix the stream.

12. Launch PRAYCG2.0 runner

The safest current workflow is through PsychoPy Coder.

1. In Control Center, open **Runner / PsychoPy**.
2. Select launch mode:

Open PsychoPy Coder + runner file/folder

3. Click **Launch PRAYCG2.0 Runner Using Selected Mode**.
4. PsychoPy Coder should open.
5. Open the runner file if it did not open automatically.
6. Press **Run** inside PsychoPy Coder.

In the runner media-selection GUI, confirm:

participant_id
session_id
run_label
control_video
target_video
override_video
cue_schedule_json
predeclared_anchor_file
baseline_seconds
washout_seconds
final_reflection_baseline enabled
per_phase_confound_reports enabled
override_rsm_report enabled
channel_map confirmed and locked

The runner should record hashes for the selected media and should report whether Target and Override hash-match.

13. PRAYCG2.0 run sequence

A standard PRAYCG2.0 run proceeds approximately as:

Setup / media selection / marker stream online

Baseline 1

Control instructions

Control video

Washout 1

Control self-report / confound report

Target instructions

Target video

Washout 2

Target self-report / confound report

Override instructions

Override video with analytic task

Washout 3

Override self-report / confound report

Override task report

Final reflection baseline

Final comparative report

Protocol end

The final reflection baseline is important for MRED-Resolution, NUPI, DGA, and after-state interpretation.

14. During the run

Do:

- ☐ Stay still during baselines and videos.
- ☐ Avoid jaw clenching.
- ☐ Avoid unnecessary blinking/squinting.
- ☐ Keep eyes open unless instructed otherwise.
- ☐ Let Target land naturally.
- ☐ During Override, perform the cue task as instructed.
- ☐ Report confounds honestly.

Do not:

- ☐ stop LabRecorder during washout
 - ☐ restart a streamer without noting it
 - ☐ ignore major audio problems
 - ☐ treat a confounded run as confirmation-grade
-

15. After the run

After the protocol ends:

1. Stop LabRecorder.
2. Confirm XDF exists.
3. Close PsychoPy.
4. Stop BrainFlow.
5. Stop Polar.
6. Stop Vernier.
7. Copy runner logs into the run folder.
8. Copy or snapshot MediaPrep QC folder and anchor file.

Recommended run folder:

```
C:\PRAYCG\runs\<run_id>\
acquisition\
  *.xdf
runner_logs\
  *_events.csv
  *_events.json
  *_run_config_media_selection.json
  *_channel_map.csv
  *_core_report.json
  *_confound_report.json
  *_override_task_report.json
  *_final_master_report.json
inputs_snapshot\
  cue_schedule.json
  predeclared_anchor_file.json
  media_manifest.json
  stimulusfingerprint_cet_eet_manifest.json
notes\
  operator_notes.txt
```

16. Find template files

Templates may live in the Master Suite, anchor packs, or project docs.

MRED familiarity CSV

Purpose:

Manual covariate file describing prior exposure, familiarity, autobiographical resonance, new meaning today, and related factors.

Usually located in:

tools\MasterComprehensiveSuite_v1_5_6_CURRENT\templates\

or in a stimulus-specific anchor pack.

If missing, create it manually with columns such as:

anchor_id,prior_exposure_0_9,scene_remembered_0_9,beat_anticipated_0_9,new_insight_0_9,current_life_connection_0_9,notes

MRED scene map CSV

Purpose:

Maps anchor IDs to rendered times, labels, predicted roles, and claim level.

Common columns:

anchor_id,rendered_time_sec,scene_label,predicted_mred_quadrant,a_mred_role,primary_endpoint_use,claim_level,notes

For blind runs, use neutral labels until after acquisition.

Annotation CSV

Purpose:

Bridge file for visual review, event overlays, and report generation.

Common columns:

anchor_id,condition,start_sec,end_sec,time_sec,label,claim_level,category,notes

Feature table CSV

Purpose:

Main time-resolved physiology feature table produced by the Master Comprehensive Analysis Suite.

It is not generated by MediaPrep.

It usually appears after analysis here:

C:

\PRAYCG\analysis\<run_id>_MasterComprehensive\tables*_time_resolved_feature_frame.csv

17. Run the Master Comprehensive Analysis Suite

In Control Center:

1. Go to **Analysis**.
2. Select or scan the run folder.
3. Confirm the required files are found.
4. Click **Run Master Comprehensive Analysis Suite**.

Typical required inputs:

XDF file

PRAYCG events JSON or CSV

run_config_media_selection.json

channel_map.csv

cue_schedule.json

predeclared_anchor_file.json

StimulusFingerprint QC folder

Optional but recommended inputs:

MRED familiarity CSV

MRED scene map CSV

annotation CSV

operator notes

Expected outputs:

tables*_time_resolved_feature_frame.csv

tables*_amred_anchor_endpoint_table.csv

tables*_mred_peak_resolution_anchor_table.csv

tables*_tti_global_summary.csv

tables*_nupi_summary.csv

tables*_dga_branch_gate_table.csv

reports*.md

reports*.txt

figures*.png

18. Run the Offline Interpreter

The offline interpreter reads the analysis output folder and writes a plain-English report.

In Control Center:

Analysis -> Launch Offline Interpreter

Select:

C:\PRAYCG\analysis\<run_id>_MasterComprehensive\

Expected outputs:

reports\offline_interpretation\offline_interpretive_report.md

reports\offline_interpretation\offline_interpretive_report.txt

reports\offline_interpretation\offline_interpretive_report_summary.json

This is a rule-based local interpreter, not an AI model. It cannot replace expert review, but it gives users a consistent first-pass explanation.

19. Run the Visualizer

The visualizer creates a synchronized video review artifact.

Inputs may include:

stimulus_video.mp4

XDF or feature table CSV

event overlay CSVs

analysis output folder

selected condition: Control, Target, or Override

render mode: Public/TSC or Lab Diagnostic

Use Public/TSC mode for clean presentation.

Use Lab Diagnostic mode for internal debugging and more detailed overlays.

Visualizer output is for review only. It does not certify A-MRED, MRED, DGA, NUPI, OSM, or any mechanism.

20. Interpret the run in the correct order

Do not start with the most exciting graph. Start with QC.

Recommended interpretation order:

1. Did LabRecorder record the required streams?
2. Did the runner log the correct Control, Target, Override, cue schedule, and anchor file?
3. Did Target and Override hash-match?
4. Did ALS timing pass, if used?
5. Did EEG sample rate and stream continuity pass?
6. Did confound reports show major audio/video/noise issues?
7. Did Control audio QC pass?

8. Did StimulusFingerprint / CET-R exist?
9. Did artifact gates pass?
10. Did A-MRED or MRED-Peak/Resolution pass?
11. Do secondary modules agree or disagree?
12. Does self-report support, constrain, or contradict physiology?

Confirmation-grade runs need clean QC. A powerful subjective experience with poor audio or broken timing is still scientifically limited.

21. Module tiers in plain English

Boxed primary path

Timing/QC

+ StimulusFingerprint/CET-R

+ artifact/confound gates

-> A-MRED / MRED-Peak / MRED-Resolution

This is the main future endpoint pathway.

Secondary interpretation

NIP / BIT / CII / IAQ

TTI

NUPI

DGA

Baseline1 vs Baseline2

These help describe what kind of response occurred.

Exploratory convergence

KHT-topo

NAST

EET

MRED-ITP / ACG / OCU

OCM025 / RSM / CVB / SquintProxy

LSO / Subtitle Override when applicable

These are useful but should not be treated as standalone confirmation.

22. Claim labels

Use these labels consistently:

Hardware shakedown:

stream/timing/software test only.

Pilot:

useful run but missing strict QC, frame-locked anchors, or clean controls.

Runner-registered estimated-anchor pilot:

anchor file loaded, but times estimated rather than frame-verified.

Frame-verified locked-anchor attempt:

anchor times verified on final rendered Target MP4 before acquisition.

Confirmation-grade candidate:

locked anchors + clean timing + clean streams + clean confounds + endpoint pass.

Do not call anything confirmation-grade unless the QC supports it.

23. Common workflow failure modes

Target and Override do not hash-match

The Target-vs-Override comparison is weakened. Regenerate MediaPrep.

Control audio contains recognizable words

Control is invalid or pilot-only. Regenerate Control or document failure.

ALS pulse not detected

Check holder placement, safe-fit scaling, shroud, pulse duration, and AUX channel selection.

Run the ALS barcode test.

EEG stream has sample-rate drops or large gaps

Use the run as pilot-only or reconstruct carefully if status markers support it. Do not treat timing-sensitive endpoints as clean.

Audio confound differs across branches

Target-vs-Override interpretation is compromised. This was the key lesson of the Eternal Sunshine rain-confounded run.

Runner only works from PsychoPy Coder

Use PsychoPy Coder. Do not force direct Python launch.

24. Public upload boundaries

For GitHub:

OK:

- code
- docs
- templates
- synthetic demo fixtures
- public-safe derived tables
- public-safe reports

Do not upload:

- copyrighted video clips
- raw private XDFs
- unredacted self-report
- private participant information
- secret tokens or API keys

For OSF:

OK:

- frozen release zips
- public-safe example data
- public-safe pilot reports
- documentation packs
- no-copyright demo media

Always label pilot data as pilot data.

25. Day-of-run checklist

- ☐ Stimulus legally procured and locally stored.
- ☐ MediaPrep complete.
- ☐ Target video found.
- ☐ Override video found.
- ☐ Control video found.
- ☐ Cue schedule JSON found.
- ☐ Target/Override hash-match confirmed.
- ☐ Control audio QC passed.
- ☐ Visual/watermark QC passed.
- ☐ Anchor file selected and claim level known.
- ☐ OpenBCI board tested.
- ☐ Cap fitted.
- ☐ Polar stream tested.
- ☐ Vernier stream tested.
- ☐ BrainFlow stream tested.
- ☐ ALS pulse tested if used.

- ☐ LabRecorder sees streams.
 - ☐ PsychoPy Coder can launch runner.
 - ☐ Quiet room / audio conditions acceptable.
 - ☐ Operator notes file ready.
-

26. Minimal successful run definition

A minimal successful pilot run has:

- ☐ XDF recorded.
- ☐ PRAYCG events JSON/CSV saved.
- ☐ Control, Target, Override completed.
- ☐ Self-reports completed.
- ☐ Analysis suite can generate a feature table.
- ☐ Offline interpreter can generate a report.

A confirmation-grade candidate additionally requires:

- ☐ frame-verified locked anchors
 - ☐ clean timing markers
 - ☐ clean ALS if used
 - ☐ target/override hash match
 - ☐ control audio QC pass
 - ☐ no major branch-specific confound
 - ☐ artifact gates pass
 - ☐ StimulusFingerprint/CET-R support
 - ☐ A-MRED or MRED-Peak/Resolution endpoint pass
-

27. Final rule

The PRAYCG pipeline is powerful because it is layered. Do not skip the layers.

Stimulus QC protects the meaning claim.

Timing QC protects the event claim.

Artifact QC protects the EEG claim.

Confound reports protect the interpretation.

Self-report constrains physiology, but does not prove mechanism.

Analysis outputs are evidence, not verdicts.